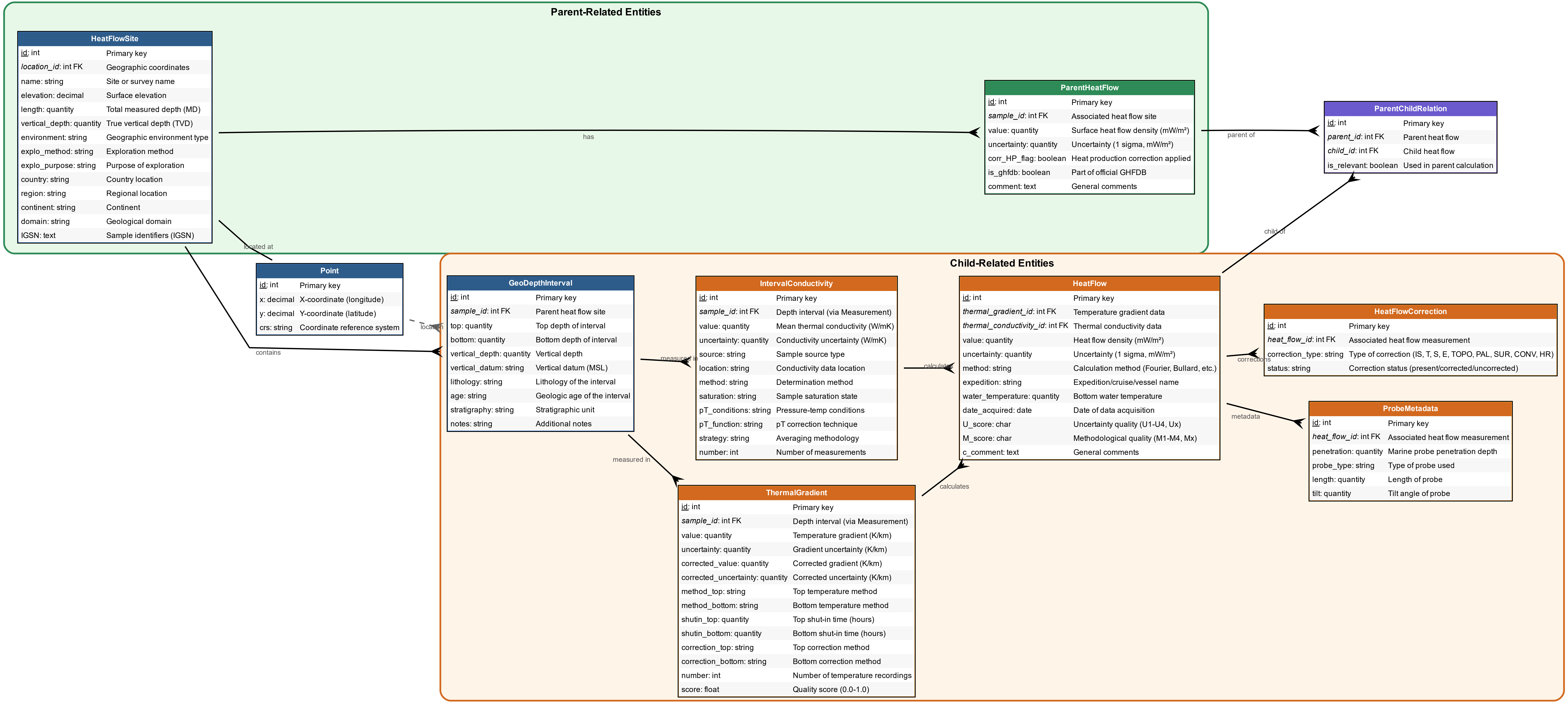
Parent-Related Entities
HeatFlowSite
id: int Primary key
location_id: int FK Geographic coordinates
name: string Site or survey name
elevation: decimal Surface elevation
length: quantity Total measured depth (MD)
vertical_depth: quantity True vertical depth (TVD)
environment: string Geographic environment type
explo_method: string Exploration method
explo_purpose: string Purpose of exploration
country: string Country location
region: string Regional location
continent: string Continent
domain: string Geological domain
IGSN: text Sample identifiers (IGSN)
has
ParentHeatFlow
id: int Primary key
sample_id: int FK Associated heat flow site
value: quantity Surface heat flow density (mW/m²)
uncertainty: quantity Uncertainty (1 sigma, mW/m²)
corr_HP_flag: boolean Heat production correction applied
is_ghfdb: boolean Part of official GHFDB
comment: text General comments
parent of
ParentChildRelation
id: int Primary key
parent_id: int FK Parent heat flow
child_id: int FK Child heat flow
is_relevant: boolean Used in parent calculation
child of
located at
Point
id: int Primary key
x: decimal X-coordinate (longitude)
y: decimal Y-coordinate (latitude)
crs: string Coordinate reference system
location
contains
Child-Related Entities
GeoDepthInterval
id: int Primary key
sample_id: int FK Parent heat flow site
top: quantity Top depth of interval
bottom: quantity Bottom depth of interval
vertical_depth: quantity Vertical depth
vertical_datum: string Vertical datum (MSL)
lithology: string Lithology of the interval
age: string Geologic age of the interval
stratigraphy: string Stratigraphic unit
notes: string Additional notes
measured in
IntervalConductivity
id: int Primary key
sample_id: int FK Depth interval (via Measurement)
value: quantity Mean thermal conductivity (W/mK)
uncertainty: quantity Conductivity uncertainty (W/mK)
source: string Sample source type
location: string Conductivity data location
method: string Determination method
saturation: string Sample saturation state
pT_conditions: string Pressure-temp conditions
pT_function: string pT correction technique
strategy: string Averaging methodology
number: int Number of measurements
calculates
HeatFlow
id: int Primary key
thermal_gradient_id: int FK Temperature gradient data
thermal_conductivity_id: int FK Thermal conductivity data
value: quantity Heat flow density (mW/m²)
uncertainty: quantity Uncertainty (1 sigma, mW/m²)
method: string Calculation method (Fourier, Bullard, etc.)
expedition: string Expedition/cruise/vessel name
water_temperature: quantity Bottom water temperature
date_acquired: date Date of data acquisition
U_score: char Uncertainty quality (U1-U4, Ux)
M_score: char Methodological quality (M1-M4, Mx)
c_comment: text General comments
corrections
HeatFlowCorrection
id: int Primary key
heat_flow_id: int FK Associated heat flow measurement
correction_type: string Type of correction (IS, T, S, E, TOPO, PAL, SUR, CONV, HR)
status: string Correction status (present/corrected/uncorrected)
metadata
ProbeMetadata
id: int Primary key
heat_flow_id: int FK Associated heat flow measurement
penetration: quantity Marine probe penetration depth
probe_type: string Type of probe used
length: quantity Length of probe
tilt: quantity Tilt angle of probe
measured in
calculates
ThermalGradient
id: int Primary key
sample_id: int FK Depth interval (via Measurement)
value: quantity Temperature gradient (K/km)
uncertainty: quantity Gradient uncertainty (K/km)
corrected_value: quantity Corrected gradient (K/km)
corrected_uncertainty: quantity Corrected uncertainty (K/km)
method_top: string Top temperature method
method_bottom: string Bottom temperature method
shutin_top: quantity Top shut-in time (hours)
shutin_bottom: quantity Bottom shut-in time (hours)
correction_top: string Top correction method
correction_bottom: string Bottom correction method
number: int Number of temperature recordings
score: float Quality score (0.0-1.0)